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CAMBRIDGE
INTERNATIONAL EXAMINATIONS

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INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 80

SYLLABUS/COMPONENT: 0625/03

PHYSICS
Paper 3 (Extended)

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1	**(a) (i)**	7(.0 s)	**A1**	
	(ii)	PQ or 0 – 2s or other correct description	**A1**	
		distance = av. speed x time or area under graph	**C1**	
		distance 11 x 2 m= 22 m	**A1**	**4**
	(b) (i)	deceleration (now) uniform (test 2)	**B1**	
		slower/lower (average) value/value between that of PQ and QR/takes longer (or values) time to come to rest.	**B1**	
	(ii)	deceleration = change in speed/time or 15/8	**C1**	
		value = 1.9 m/s^2	**A1**	**4**
	(c) (i)	graph shows constant acceleration	**B1**	
		force = ma (and m is also constant) so force is constant	**B1**	
	(ii)	towards the centre of the motion/circle	**A1**	**3**
				[11]
2	**(a)**	pressure = depth x g x density of water	**C1**	
		pressure = 50 x 10 x 1000	**C1**	
		so value is 500 000 Pa or N/m^2	**A1**	**3**
	(b)	force = pressure x area in any form	**C1**	
		force = 500 000 x 0.15 x 0.07	**C1**	
		force = 5250 N	**A1**	**3**
				[6]
3	**(a)**	one slightly nearer the centre than the other	**C1**	
		20 kg is the nearer one to the pivot	**A1**	**2**
	(b)	Clockwise moments = anticlockwise moments (about point/pivot)	**A1**	**1**
		(accept opposite directions and equal)		
	(c)	18x2.5=20xB	**C1**	
		distance = 2.25(m)	**A1**	**2**
				[5]
4	**(a)**	Some have extra/more energy than others	**B1**	
		most energetic leave surface/ break liquid bonds etc	**B2**	**M2**
	(b)	evaporation occurs strictly at the surface/at all temperature	**B1**	
		boiling occurs throughout liquid/ at one temperature (at normal at. pr.)/100°C	**B1**	**2**
	(c)	energy supplied = Wt /60 x 120	**C1**	
		sp.latent heat = energy/mass evaporated or 60 x 120/3.2	**C1**	
		value is 2250 J/g	**A1**	**3**
				[7]
5	**(a) (i)**	nitrogen	**M1**	
	(ii)	copper-solid-molecules very tightly bonded together so separate little	**B1**	
		water – liquid – molecules less tightly bonded/still small separation	**B1**	
		nitrogen – gas – molecules “free” and not bonded so separate most	**B1**	**M3**
		(N.B. accept 2 bonding statements for 2 marks. 1 separation statement for 1 mark)		

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	(b) (i)	size of movement/change in length of liquid column per degree	B1		
	(ii)	change in length (of liquid column) same for all degrees	B1		
					[5]
6	(a)	3 more roughly circular	B1		
		all drawn clearly circular, stop (well) clear of barrier and centred on slit	B1		
		wavelength constant throughout, both sides of barrier	B1		3
	(b)	wavelength – speed/frequency in any form	C1		
		values substituted correctly	C1		
		answer 6 x 10 m	A1		3
					[6]
7	(a)	two dots, marked F, each 5.0 cm from the lens	A2		2
	(b)	each correct ray one mark	M2		2
	(c)	correct image, labeled I	A1		1
	(d)	rays pass along the axis undeviated/object distance same for all object/rays meet at same distance on image/image distance same for all image	B1		1
	(e)	magnifying glass/eyepiece of telescope or microscope	B1		1
					[7]
8	(a) (i)	0-6 (V) positive and negative	A1		
	(ii)	all waves roughly 6V amplitude	B1		
		3 waves approx. one wave every 0.1 s	B1		3
	(b)	any mention of magnetic field	B1		
		coils (forced to) cut magnetic field	B1		
		includes e.m.f./voltage/current in the coils	B1		
		as in Fleming's R.H. rule	B1		M3
	(c)	mechanical energy/work (in)/kinetic energy	B1		
		electrical (out) (+ heat) (ignore sound)	B1		2
					[8]
9	(a) (i)	regular (but)/not normal (sine) wave/several waves added together etc.	B1		
	(ii)	1.6(V)	A1		
	(iii)	connect known voltage to Y plates (without any changes to C.R.O.)	B1		
		read off against screen values	B1		4
	(b) (i)	6.1 (cm) (accept 6 or any value in range 6.0 to 6.2)	A1		
	(ii)	50 ms for 10 cm or 5 ms per cm e.c.f.	C1		
		so 6.1 x 5 ms or 31 ms	A1		
	(iii)	difference in time of runners finishing race or other timing between two closely separated events.	B1		4
					[8]

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10	**(a)**	current = power/voltage or 150/12	**C1**	
		value is 12.5 A	**A1**	
	(b) **(i)**	sum of currents at junction = current after junction/12.5 A = 5.0 A + I	**C1**	
		value is 7.5 A	**A1**	
	(ii)	power = VI or is 7.5 x 12 e.c.f from **(i)**	**C1**	
		value is 90 W	**A1**	
	(iii)	resistance = voltage/current or 12/7.5 e.c.f. from **(i)** but not from **(a)**	**C1**	
		value is 1.6Ω	**A1**	**6**
				[8]
11	**(a)**	top line correct, need 24 and 0	**B1**	
		bottom line correct, need 12 and –1 (accept β or e for electron	**B1**	**2**
	(b)	particles take curved path (accept from diagram)	**B1**	
		move between the poles at right angles to lines of force	**B1**	
		move out of paper	**B1**	**3**
	(c) **(i)**	use detector to pick up <u>radiation</u> (from isotope at points on/in body etc.)	**B1**	
		high count where circulation good or v.v. explained	**B1**	
	(ii)	alpha particles all absorbed, none detected		
		beta particles may be largely absorbed, not penetrative enough		
		gamma rays reach detector/leave body any two	**B2**	**4**
				[9]
				TOTAL 80